

AURA Forge Architecture Write-Up

Governed adaptive engineering workflow for FitGPT

AURA Forge is a capstone engineering-governance system around FitGPT. It classifies a change by risk, selects the minimum justified route, enforces least-privilege MCP boundaries, and preserves evidence through CI artifacts and a Change Passport.

1. Workflow Overview

Change request	Scope and safety precheck	Deterministic risk classifier
Adaptive route	Bounded roles/tools	Policy and evaluation gates
Human checkpoints when required	Change Passport	Audit trail / grader evidence

The workflow starts with repository evidence rather than a model decision. The classifier assigns LOW, MEDIUM, or HIGH risk from paths and metadata. The router then chooses a frozen route ID and role sequence. Evidence producers generate policy, evaluation, approval, metrics, and CI outputs for the Passport.

2. Agent Roles And Human Checkpoints

Tier	Route	Model roles	Human checkpoints	Governance meaning
LOW	Implementer -> Reviewer -> deterministic verification -> terminal readiness	2	0	Use lightweight autonomy for non-executable or mechanically verifiable changes.
MEDIUM	Implementer -> Reviewer -> Tester -> final approval -> terminal readiness	3	1	Keep implementation, independent review, test evidence, and final readiness approval for bounded executable work.
HIGH	Planner -> plan approval -> Implementer -> Reviewer -> Tester -> policy/eval gates -> final approval -> Project Manager	5	2	Preserve full governance for security, MCP, CI, schema, credential, policy, and eval-sensitive changes.

3. Deterministic Steps

- Risk classification is deterministic and model-free (`aura-risk-v1`).
- Route selection is deterministic and frozen (`aura-router-v1`).
- Pipeline integrity checks parse workflow YAML and reject weakened permanent gates.
- Change Passport generation reads producer files and hashes; missing producers fail instead of fabricating fields.
- Stable `DATABASE\_URL` consistency checking was converted from an agentic review to deterministic code with \$0 model cost.

4. MCP Boundaries And Least Privilege

Role	Allowed course-tool profile	Denied boundary
Planner	Read-only evidence for a bounded plan.	No writes, tests, tracker, broad shell, or web search.
Implementer	Bounded file read/write only to approved paths.	No task tracker, permission mutation, production access, or self-approval.
Reviewer	Read-only review of diff and evidence.	No writes or tracker updates.
Tester	Focused bounded test execution.	No repairs, broad shell, or tracker updates.
Project Manager	Task tracker only after final approval.	No repository read/write, shell, tests, or web search.

5. Data Flow, Storage, And Retrieval

- Source evidence lives in committed repository files under ``docs/capstone/`, `docs/adr/`, `eval/`, and `.github/workflows/ci.yml`.`
- Generated readiness evidence is represented by the tracked Change Passport JSON: ``docs/capstone/evidence/change-passport-AF-HIGH-001.json`.`
- Local ignored eval artifacts are not required uploads; submitted PDFs summarize only sanitized, committed evidence.
- No production database, production secret, or real user/customer data is required for deterministic verification.

6. CI/CD Integration And Governance Enforcement

Push to capstone/aura-forge	change-classifier	policy-tests
advisory-review	evaluation-gate	pipeline-integrity
audit-trail	sanitized artifacts	Change Passport evidence

- Permanent gates are deterministic and blocking: policy tests, evaluation gate, and pipeline integrity.
- Advisory AI review is non-blocking and safely emits ``SKIPPED - AI SECRET UNAVAILABLE`` when optional credentials are absent.
- Governance denial ``GO-20260811-001`` proves ``implementer -> task_tracker`` is denied with no external state change and \$0 model cost.

7. Explicit Non-Claims

- No Render or Vercel production deployment is claimed.
- No live FitGPT mutation, production database write, or real user data is part of this capstone.
- Network egress is bounded by course verification practices but is not claimed airtight.

Source Evidence

Primary sources: docs/capstone/final-architecture.md, docs/capstone/adaptive-routing.md, docs/routing-and-tool-grant-map.md, docs/capstone/governance-overreach-demo.md, docs/capstone/evidence/change-passport-AF-HIGH-001.json, .github/workflows/ci.yml.